Python Engineering - Class 06

Dictionaries and Sets | EX14 - Simulation Configuration

Class objective: Organize parameters and results with descriptive keys and validate required parameter sets.

This document is the base exercise guide. It is not a full solution to copy: it organizes the problem, implementation steps, reference values and required validation.

Path

```
01_python_bootcamp/
`-- week_02_data_structures/class_06_dictionary_sets/
    |-- notes/
    `-- exercices/
```

Required workflow

```
PROBLEM
INPUTS
OUTPUTS
EQUATIONS
SPECIAL CASES
DO I NEED ITERATION?
ALGORITHM
VALIDATION
```

Before coding, write these eight sections. The goal is to separate problem reasoning from syntax.

EX14 - Simulation Configuration with dict and set

Problem: Represent vertical-simulation configuration/results with dictionaries and verify required keys using sets.

Reference inputs: config keys such as h0,v0,t0,g,dt

Equations / relationship: Same vertical-motion model as previous classes

Expected check: required_keys - provided_keys should be empty for a complete configuration.

Implementation guide

- 1. Create config dict.
- 2. Define required_keys set.
- 3. Compare available keys.
- 4. Access values by name instead of index.
- 5. Return labelled results in a dict.

Before running

Predict sign, order of magnitude, data type and any relevant shape. Then compare the prediction with the actual output.

Final validation

Answer these questions as comments at the end of the exercise. The goal is not memorized wording but demonstrating that you can explain the code flow.

- Difference between list and dict.
- What are keys and values?
- What does key in config check?
- What does required - provided mean?
- Why does a set remove duplicates?
- When is returning a dict clearer than a tuple?

Completion criterion

The class is complete when the code runs, the reference checks make sense, you can explain every block without reading a solution, and you can modify at least one input while predicting the expected effect.